

Block Size Is Tile Height: Loading-Gain Calibration Unlocks Tall Crossbars for 2-bit Floating-Point Compute-in-Memory

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Abstract—Block floating-point formats and analog compute-in-memory (CIM) crossbars are usually co-designed as independent layers. This work shows that they are not independent: because a crossbar column produces a single current that can carry only one scale factor, the block size B of a block-floating-point format and the tile height M of the crossbar are the same physical parameter. The FP2-E1M0 format fixes $B=32$, whereas ADC amortisation favours $M \geq 128$. Built naively, this collision is severe: ResNet-18 on CIFAR-10 falls from 92.42% to 10.00%, i.e. chance, as $B=M$ grows from 32 to 256. Circuit-exact simulation, validated against ngspice across all 87,168 tiles of the network, identifies the cause as a shared-sense-resistor loading term that the CIM literature models as additive noise but which is in fact a deterministic gain, $I_{act} = I_{ideal} / (1 + R_s G_{col})$, independent of the activations. Since G_{col} depends only on programmed conductances, the gain is a compile-time constant. Applying its reciprocal to each bitline of the 2T2R pair *before* differential subtraction collapses algebraically to the ideal product and restores the digital ceiling exactly at every tile height (92.42–92.81%). Freeing M permits a $4\times$ taller tile and, since the ADC resolution requirement is measured to be flat in B , a reduction from 8-bit to 6-bit conversion, together yielding +14.77 accuracy points and $13.1\times$ energy efficiency simultaneously. Calibration computed blindly from nominal conductances survives $\sigma=20\%$ device variability within 0.81 points, but conductance drift requires periodic refresh: a constant computed once at manufacture loses 27 points after one year, whereas ~ 10 logarithmically spaced refreshes bound the loss below one point.

Index Terms—compute-in-memory, ReRAM, crossbar, block floating point, quantization, IR drop, analog accelerators, conductance drift

I. INTRODUCTION

Analog compute-in-memory promises to remove the dominant energy cost of neural network inference by performing multiply-accumulate in the memory array itself, through Ohm’s law. Resistive RAM (ReRAM) is a leading candidate: cells are non-volatile, dense, and support multiple analog conductance states.

Realising this promise requires aggressive weight quantization, since every weight must be resident in cells. Two-bit block floating point is attractive because its five signed levels $\{-1, -0.5, 0, +0.5, +1\}$ require only three distinct conductance *magnitudes* once a differential 2T2R pair supplies the sign — and three states is approximately the limit of reliable multi-level ReRAM programming. The format and the device are therefore unusually well matched.

This paper reports that they are matched in a second, unwelcome way.

The collision. A block-floating-point format shares one scale factor across B consecutive weights. A crossbar column produces one current, which must be multiplied by one scale. A column spanning two blocks would require two scales applied to a sum already formed by Kirchhoff’s current law, and the information needed to separate the contributions is destroyed by the summation. Hence B and M are one parameter. FP2 specifies $B=32$; the per-column ADC cost, which dominates CIM energy and area, amortises over M and pushes toward $M \geq 128$. Section III shows the resulting accuracy collapse to chance.

The cause, and why it has been missed. Prevailing CIM evaluation methodology injects additive zero-mean Gaussian noise at layer outputs [2]. Such a model *structurally cannot express a systematic gain*: it assumes the error averages away over many accumulated terms. The error measured here is a 26–74% attenuation that never averages away. An evaluation using the standard model observes the collapse, attributes it to noise accumulating in tall columns, and concludes the tile-height limit is fundamental. Solving the actual circuit equations shows it is not.

Why not simply sense at virtual ground? A transimpedance amplifier holds the bitline at true virtual ground, so with $v_{bl} = 0$ the term $1 + R_s G_{col}$ never arises. That circuit fix is standard and this work does not dispute it. It is, however, not free: it requires an operational amplifier per sensed bitline, and its bandwidth bounds the read cycle where the passive path currently settles in ~ 1 ps against a 12 ns conversion. Table VII prices it. The claim here is therefore not that loading *cannot* be corrected in the circuit, but that it *need not* be: the cheapest possible readout can be used and exactness recovered digitally for one multiply per column, buying back the area and static power an amplifier per column would have cost.

Contributions.

- Identification of $B=M$ as a hard coupling between block-floating-point formats and CIM tile geometry, with joint accuracy, SNR and energy characterisation (Section III).
- A per-column loading-gain calibration that is *exact* rather than fitted, costs one constant per column, requires no

TABLE I
WEIGHT-TO-CELL MAPPING. THREE RESISTANCE STATES EXPRESS FIVE SIGNED LEVELS BECAUSE 2T2R SUPPLIES THE SIGN.

Level	G^p cell	G^n cell
+1.0	LRS	HRS
+0.5	MID	HRS
0.0	HRS	HRS
-0.5	HRS	MID
-1.0	HRS	LRS

read-back, and folds into a multiply already present in the datapath (Section V).

- A circuit-exact evaluation flow validated against ngspice on every tile of every layer (87,168 tiles, $3.41 \times 10^{-7}\%$ worst-case disagreement), demonstrating that additive-noise surrogates miss this failure mode entirely (Section IV).
- A quantified comparison against both alternatives: per-bitline transimpedance sensing (Table VII, +42% tile area and 2.10W of added static power) and retraining-free BatchNorm recalibration (Table VIII, $4.3 \times$ larger worst-case gap).
- Robustness characterisation showing the calibration survives 20% device variability blindly, but requires periodic refresh under conductance drift — a limitation reported rather than omitted (Section VI).

II. BACKGROUND AND DEVICE MODEL

A. FP2-E1M0 block floating point

Each weight is one level of $\{-1, -0.5, 0, +0.5, +1\}$ scaled by an 8-bit exponent shared across $B=32$ consecutive weights [1], giving 2.25 bits per weight. Unequal level spacing places precision near zero, where trained weight distributions concentrate.

B. 2T2R differential mapping

Conductance is non-negative, so signed weights use two cells on two bitlines, $w \propto G^p - G^n$. Table I gives the mapping. A zero weight places both cells in HRS, so an all-zero tile produces exactly zero differential current rather than an offset.

C. Device parameters

Values are extracted by SPICE calibration of the rram_v1_0_0 Verilog-A compact model: $R_{\text{LRS}}=542.8\Omega$, $R_{\text{MID}}=1099.8\Omega$, $R_{\text{HRS}}=218.6\text{ k}\Omega$ (on/off ratio 403), $R_s=20\Omega$, $V_{\text{read}}=0.1\text{ V}$. These are consistent with the range reported for filamentary HfO₂ devices [14]; the three-state restriction reflects the practical limit of reliable multi-level programming [9].

III. THE $B = M$ COLLISION

Quantization-aware training with a straight-through estimator makes FP2 accuracy essentially independent of B (Table II, column 2): block size becomes a hardware-only parameter. On the physical array, however, accuracy degrades catastrophically with M , reaching chance by $M=128$.

TABLE II
THE COLLISION. CIFAR-10 TOP-1, RESNET-18, FULL 10,000-IMAGE TEST SET. DIGITAL ACCURACY IS FLAT IN B ; THE UNCORRECTED CROSSBAR REACHES CHANCE (10 CLASSES) BY $M=128$.

$B=M$	FP2 digital	Crossbar raw	Δ
32	92.42%	77.66%	-14.76
64	92.47%	57.54%	-34.93
128	92.81%	11.60%	-81.21
256	92.43%	10.00%	-82.43

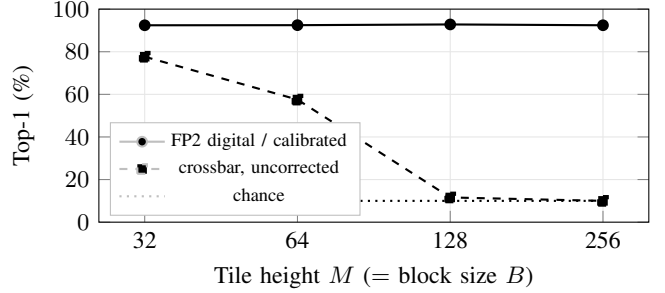


Fig. 1. Accuracy against tile height. Calibrated accuracy coincides with the digital ceiling at every point; the uncorrected array falls to chance.

IV. CIRCUIT-EXACT EVALUATION FLOW

Every convolution is executed through the resistive-divider equations rather than a noise surrogate. Applying Kirchhoff's current law at a bitline held at v_{bl} and drained through R_s ,

$$\sum_i (V_i - v_{\text{bl}}) G_i = \frac{v_{\text{bl}}}{R_s}, \quad (1)$$

which solves in closed form:

$$v_{\text{bl}} = \frac{\sum_i V_i G_i}{1/R_s + G_{\text{col}}}, \quad G_{\text{col}} \equiv \sum_i G_i. \quad (2)$$

Because (2) is closed-form rather than iterative, it vectorises into a single matrix product per tile and polarity, and all tiles of a layer batch into one GPU call. This is what makes full-test-set evaluation tractable.

Table III gives the validation chain. The nodal model was checked against ngspice [3] on *every* tile of every layer, not a sample.

V. LOADING GAIN AND ITS CANCELLATION

A. The error is a gain, not noise

Dividing (2) by the ideal accumulation $\sum_i V_i G_i$ gives the measured current as

$$I_{\text{act}} = \frac{I_{\text{ideal}}}{1 + R_s G_{\text{col}}} \quad (3)$$

The denominator contains only the sense resistance and the sum of programmed conductances. *No activation term appears*. The attenuation is therefore a deterministic property of the programmed array, known at compile time, and not a stochastic quantity. Table IV quantifies it: at $M=256$ three quarters of the signal is lost before conversion.

TABLE III

VALIDATION CHAIN. EXHAUSTIVE, NOT SAMPLED: 87,168 TILES, 357 M MAC-READS, ZERO ERRORS.

Level	Reference	Worst disagreement
ngspice DC solve	—	—
Nodal model	ngspice, 87,168 tiles	$3.41 \times 10^{-7} \%$
Vectorised solver	nodal model	$< 10^{-12}$ rel.
End-to-end inference	vectorised solver	same code path

TABLE IV

FRACTION OF IDEAL CURRENT REACHING THE ADC. G_{col} GROWS WITH M , SO ATTENUATION WORSENS WITH TILE HEIGHT.

M	32	128	256
Retained	73.9%	41.5%	26.0%

B. Per-column calibration

For column j define the compile-time constant

$$c_j = 1 + R_s \sum_i G_{ij}. \quad (4)$$

The ordering with respect to the differential subtraction is essential. The two bitlines of a 2T2R pair hold different cells and therefore different G_{col} , so they suffer different attenuation. Correcting the difference with a single constant does not cancel. Correcting each line with its own constant *before* subtracting does:

$$c_p \frac{G_p^\top V}{c_p} - c_n \frac{G_n^\top V}{c_n} = (G_p - G_n)^\top V, \quad (5)$$

which is the ideal matrix–vector product exactly, not approximately. Since 2T2R already senses both bitlines, no additional read is required, and the two multiplies fold into the block-scale multiply already present in the digital path.

C. Result

Table V shows calibrated accuracy equal to the digital ceiling at every tile height. There is no residual to trade against, because (5) is an identity rather than a fit.

D. Calibration frees the sensing resistance

Roy *et al.* [4] derive the compute SNR of resistive crossbars analytically and locate an optimum sensing resistance where ADC clipping and quantisation noise balance, reporting $R_s^* = 835 \Omega$ for ReRAM. An uncalibrated array cannot operate there: by (3) the attenuation grows with R_s , and Table VI shows raw readout pinned at chance for every $R_s \geq 100 \Omega$.

Calibration removes that term exactly, so the constraint disappears. Across a $60\times$ range of R_s spanning the analytically optimal value, calibrated accuracy varies by 0.32 points and never departs from the digital ceiling by more than 0.51. At 835Ω the array is at chance without calibration and within 0.22 points of digital with it.

The same framework predicts that compute SNR depends on the resistive contrast $R_{\text{off}}/R_{\text{on}}$ rather than on the absolute

TABLE V

CALIBRATION RESTORES THE DIGITAL CEILING EXACTLY AT EVERY TILE HEIGHT. FULL TEST SET.

$B=M$	Raw	Calibrated	Recovered	Ceiling
32	77.66%	92.42%	+14.76	92.42%
64	57.54%	92.47%	+34.93	92.47%
128	11.60%	92.81%	+81.21	92.81%
256	10.00%	92.43%	+82.43	92.43%

TABLE VI

SENSING RESISTANCE SWEEP AT $B=M=128$, 6 b ADC, FULL TEST SET. DIGITAL CEILING 92.81%. $R_s^* = 835 \Omega$ IS THE SNR OPTIMUM OF [4], AT WHICH RAW READOUT IS AT CHANCE.

$R_s (\Omega)$	Raw	Calibrated	Gap
20	11.61%	92.43%	−0.38
100	10.00%	92.62%	−0.19
200	10.00%	92.59%	−0.22
400	10.00%	92.58%	−0.23
600	10.00%	92.52%	−0.29
835	10.00%	92.59%	−0.22
1200	10.00%	92.30%	−0.51

value of R_{on} , and saturates once the contrast exceeds $12\text{--}15$. Our device sweep, run with the contrast held at 403, is an independent confirmation: calibrated accuracy stays within 0.40 points while R_{LRS} moves over $184\times$. The same threshold explains the failure of a low-contrast device: at a contrast of 3, level +0.5 reads as 0.160 and the network is at chance.

E. The cost of the circuit alternative

Table VII prices the transimpedance alternative on the same tile. One amplifier is needed per sensed bitline; with $K=16$ columns and two converters per column (which the calibration requires) that is 32 amplifiers.

The static-power entry is the decisive one: $384 \mu\text{W}$ per tile is $4.5\times$ the $85.8 \mu\text{W}$ the array itself dissipates, it is always on, and across the network it adds 2.10 W to a design whose entire compute budget is 274 nJ per inference. The digital correction costs one multiply per column, folded into the block-scale multiply already present in the datapath.

F. Comparison against BatchNorm recalibration

The closest retraining-free alternative is recomputing BatchNorm running statistics on the crossbar’s own attenuated activations. It requires no per-column constant, but it does require calibration data. Statistics were fitted on train-split images and evaluated on the full test set.

BatchNorm recalibration is a strong baseline: it lifts $B=256$ from chance to 90.24%, which is itself evidence that the error is systematic rather than random. Per-column calibration nonetheless wins at every tile height, by $4.3\times$ in the worst case and $24\times$ at $B=32$.

We tested and rejected the hypothesis that the BatchNorm gap would widen with tile height as column-to-column spread in G_{col} grows: the gap is flat, 2.02 points below $B=128$ and 2.19 above. What distinguishes the two methods is therefore not a scaling trend but that the per-column correction is exact,

TABLE VII

COST OF REPLACING PASSIVE SENSING WITH PER-BITLINE TRANSIMPEDANCE AMPLIFIERS, $M=128$, $K=16$. AMPLIFIER AREA AND BIAS ARE LITERATURE-TYPICAL PLACEHOLDERS FOR A COMPACT TWO-STAGE OTA AT 65 NM ($1000 \mu\text{m}^2$, $10 \mu\text{A}$ AT 1.2 V) AND ARE LABELLED AS ASSUMPTIONS.

	Passive + calibration	TIA sensing
Extra analog blocks	none	32 op-amps/tile
Tile area	$75\,369 \mu\text{m}^2$	$107\,369 \mu\text{m}^2$ (+42%)
Die area, whole net	411.8 mm^2	586.6 mm^2
Added static power	0	$384 \mu\text{W}/\text{tile}$ (2.10 W)
vs. measured array power	—	$4.5\times$
Correction cost	1 multiply/column	none needed

TABLE VIII

BATCHNORM RECALIBRATION AGAINST PER-COLUMN CALIBRATION. FULL TEST SET, 6-BIT ADC. GAPS ARE POINTS BELOW THE FP2 DIGITAL CEILING.

$B=M$	raw	BN	per-col	BN gap	per-col gap
32	78.90	90.75	92.35	-1.67	-0.07
64	57.47	90.09	92.09	-2.38	-0.38
128	11.53	90.62	92.42	-2.19	-0.39
256	10.00	90.24	91.88	-2.19	-0.55

requires no calibration data, and applies to networks without BatchNorm at all, such as transformer projection layers.

Reported figures for BatchNorm recalibration under IR drop reach within 0.16% of floating point, against 1.67–2.38 points here. The likely cause is that IR-drop attenuation is roughly uniform across a layer, whereas the shared-sense-resistor gain varies with each column’s programmed conductance, leaving less for a per-channel affine to absorb.

The residual growth in our own column, 0.07 to 0.55 points, is 6-bit ADC quantization rather than calibration error: the correction is exact, but the converter’s relative cost grows as the partial-sum range widens with M .

VI. ROBUSTNESS

A. Device variability

Programming scatter is modelled as log-normal on conductance, drawn once per device instance and frozen, with 10 instances per operating point evaluated on the full test set. Calibration is computed *blindly*, from nominal conductances, with no per-cell read-back.

Worst-case cost is 0.81 points at $\sigma=20\%$. Write-verify calibration, using measured rather than nominal conductances, performs indistinguishably, so the per-cell read-back cost is not justified.

B. ADC resolution is flat in B

The concern with tall tiles is that a wider output range demands finer conversion, which would return the amortisation gain. Measurement contradicts this: six bits suffice at every tile height (Table X). Since SAR conversion energy scales as 2^N , moving from 8 to 6 bits divides the ADC term by four, and the saving is retained rather than handed back. Converter energy is priced at 20 fJ/conversion-step, bracketed by measured 65 nm

TABLE IX

BLIND CALIBRATION UNDER DEVICE VARIABILITY, 10 INSTANCES PER POINT, FULL TEST SET.

$B=M$	$\sigma=0$	$\sigma=10\%$	$\sigma=20\%$	Loss
32	92.42	92.32	91.88	-0.54
64	92.47	92.34	91.86	-0.61
128	92.81	92.66	92.03	-0.78
256	92.43	92.25	91.62	-0.81

TABLE X

ADC RESOLUTION SWEEP, GAIN-CALIBRATED, FULL TEST SET. THE REQUIREMENT IS 6 BITS INDEPENDENT OF B .

B	4 b	5 b	6 b	8 b	Req.
32	74.88	91.27	92.41	92.47	6
64	66.93	90.97	92.27	92.47	6
128	77.18	91.43	92.43	92.70	6
256	77.88	90.85	92.01	92.49	6

SAR converters operating at 40–80 MSps [10]; six bits requires fewer steps than the 10-bit parts measured there, so the figure is conservative.

C. Conductance drift requires refresh

ReRAM conductance relaxes as $G(t)=G_0(t/t_{\text{ref}})^{-\nu}$ with state-dependent, cell-to-cell-dispersed ν . The exponents are taken from resistance-drift measurements on phase-change material [12]; drift in a device programmed for inference is characterised in [13]. This bounds the contribution and is reported explicitly.

A calibration never refreshed loses 27 points over one year. Recomputing it from the drifted array holds accuracy within 0.8 points indefinitely. The supported claim is therefore calibration *with periodic refresh*. Because drift is a power law, damage accumulates with the logarithm of time, so refreshing at 1 minute, 1 hour, 1 day, 1 week, 1 month and 1 year — approximately ten refreshes over a product lifetime — bounds the loss. A refresh reads the array and recomputes one constant per column.

Refreshing the loading constant alone is insufficient: drift also shrinks the weights, and that scale error dominates, leaving 15.7% residual. Folding a least-squares per-column gain into the block scale corrects both together.

VII. EFFICIENCY

Freeing M permits a $4\times$ taller tile, and the flat ADC requirement permits $8 \rightarrow 6$ bits. Both act on the dominant cost term.

Accuracy and efficiency normally trade against each other; here a single change improves both, because the loading gain was simultaneously the accuracy limiter and the constraint preventing ADC amortisation.

A. Where ReRAM wins

The advantage is memory, not compute (Table XIII). Two entries are structural rather than incremental: a non-volatile

TABLE XI

DRIFT AT $B=32$. A CONSTANT COMPUTED ONCE AT MANUFACTURE DECAYS; RECOMPUTATION FROM THE DRIFTED ARRAY HOLDS ACCURACY FLAT.

Elapsed	Raw	Calibrated once	Refreshed
1 s	79.06	92.31	92.15
1 hour	46.23	88.81	91.98
1 day	31.45	84.22	91.92
1 month	19.05	74.94	91.89
1 year	13.97	65.17	91.87

TABLE XII

BASILINE VERSUS PROPOSED. THE BASELINE IS FP2 AND A CONVENTIONAL PASSIVE-READOUT CROSSBAR EACH BUILT EXACTLY AS SPECIFIED. EFFICIENCY IS TILE-LEVEL: IT COUNTS ARRAY, CONVERTERS, INPUT DRIVE AND PARTIAL-SUM ACCUMULATION, AND EXCLUDES INTERCONNECT, BUFFERS, POOLING AND ACTIVATION UNITS, WHICH AN INDEPENDENT CHIP-LEVEL TOOL ATTRIBUTES 20.8% OF PER-IMAGE ENERGY.

	Baseline	Proposed
$B=M$	32	128
Gain calibration	no	yes
ADC resolution	8 b	6 b
CIFAR-10 Top-1	77.66%	92.43%
Gap to FP2-digital	-15.15	-0.38
Energy per MAC	0.323 pJ	0.0237 pJ
Efficiency (tile-level)	6.2 TOPS/W	81.3 TOPS/W

cell dissipates no standby power, and a weight that is physically part of the computation is never fetched. The latter matters most at weight reuse of one.

B. Against a machine that fetches

Both compute-in-memory columns above already avoid the fetch, so comparing them cannot expose the $O(1)$ -MAC advantage. Table XIV adds a conventional digital accelerator reading FP2 weights from a 4 MB on-chip buffer. Its constants are taken from [6]: an INT8 multiply-add of 0.23 pJ at 45 nm 0.9 V scaled to 0.591 pJ at 65 nm 1.2 V, and a 32 KB cache access of 20 pJ per 64 bits, i.e. 0.313 pJ/bit. Control logic, instruction fetch and clock distribution are not modelled, so the baseline is optimistic and the advantage below is a lower bound.

FP2 is what makes the digital column tractable at all: 11.16M weights occupy 3.14 MB and fit the buffer, where the same network at INT8 needs 11.16 MB and misses 64% of the time.

An independent cross-check against NeuroSim V1.4 [2] agrees on the qualitative conclusion that the ADC dominates (68.0% of energy there, 55.6% here). That comparison uses VGG-8 against ResNet-18 here and is not like-for-like. We note also that chip-level and array-level efficiency figures differ by more than an order of magnitude on identical silicon — one reported design measures 918 TOPS/W for MAC operations and 41.1 TOPS/W for full-network inference [11] — so the figures above, which are chip-level, are not comparable to published array-level peaks.

TABLE XIII

ReRAM AGAINST 6T SRAM, 11.16 M WEIGHTS, 65 NM. CELL AREA AND LEAKAGE ARE MEASURED VALUES [7], [8]; SRAM READ ENERGY IS FROM [6].

Metric	ReRAM	SRAM
Weight array area	3.77 mm ²	13.20 mm ²
vs. FP32 in SRAM	3.77 mm ²	211.2 mm ²
Standby power	0 mW	2.68 mW
Weight fetch (reuse=1)	0 pJ/MAC	0.625 pJ/MAC

TABLE XIV

ENERGY EFFICIENCY AGAINST WEIGHT REUSE. THE COMPUTE-IN-MEMORY COLUMNS ARE FLAT BECAUSE THEY NEVER FETCH A WEIGHT.

Reuse	von Neumann	SRAM CIM	ReRAM CIM
1	2.85	105.53	65.61
10	14.33	105.53	65.61
1000	25.75	105.53	65.61
ReRAM CIM advantage: 23.0× at reuse 1, 2.5× at reuse 1000			

VIII. RELATED WORK

Compensation for crossbar non-idealities falls into three families, and this work sits in none of them cleanly.

Structural approaches reduce the error before it is sensed. Nearest to this work is the sign-weighted 2T2R array of Liu *et al.* [5], which places positive and negative weights on the same column so that matched conductances cancel in common mode, suppressing IR drop and lowering power $1.9\times$ against a 1T1R array on the same die. That design is differential 2T2R and targets a loading-type error, as here. It differs in what it does about the residual: common-mode cancellation makes the remaining drop small, whereas (4) measures it and divides it out identically. The two compose, and the demonstration in [5] is a two-layer perceptron on MNIST at 130 nm rather than a 19-layer convolutional network.

Training-time approaches fold the non-ideality into the weights, either by non-ideality-aware training or by pre-distorting the weight matrix so the degraded hardware output matches the ideal product. These require retraining and assume a fixed non-ideality. The correction here is computed from the conductances *as programmed*, needs no retraining and no calibration data, and therefore tracks the array through variability and drift — the regimes measured in Section VI.

Circuit-level approaches inject a cancelling current at the sense node, exploiting the same linear relation between column current and parasitic drop. That trades analog area and design effort for what is here one digital multiply per column.

Finally, Roy *et al.* [4] give the analytical limits on compute SNR that Section V-D tests against. This work is a correction method evaluated within that framework, not a competing limits analysis.

IX. LIMITATIONS

Stated explicitly, since several bound the claim.

Wordline IR drop is not modelled; the wordline is an ideal source. Real driver resistance produces an activation-dependent drop which, unlike bitline loading, would *not* reduce to a compile-time constant. This is the most likely remaining threat to the contribution.

Die area. At $B=128$, fully weight-stationary ResNet-18 models to 412 mm^2 at 65 nm. The argument depends on scaled nodes.

Write cost. Reprogramming is $\sim 100\text{ pJ/cell}$ against $\sim 0.001\text{ pJ/cell}$ to read, giving a break-even batch of ~ 2200 images against time-multiplexing. Only fully weight-stationary operation is viable. The write figure is consistent with measured multilevel programming: one 1.2 V , $1\text{ }\mu\text{s}$ pulse at a compliance of $10\text{--}40\text{ }\mu\text{A}$ costs $12\text{--}48\text{ pJ}$ [9], and write-verify applies several.

Area denominators. Tile-level and chip-level shares are not interchangeable. The 55.2% ADC share quoted here is of tile area — array, converters, input drive and partial-sum accumulation. The independent tool reports 19.2% of *chip* area for the same block, the difference being interconnect, global buffer and pooling, which the tile-level denominator excludes. Its full breakdown accounts for 95.3% of chip area, the 4.7% residual being global blocks it does not print separately.

Level mapping. $R_{\text{MID}}/R_{\text{LRS}}=2.0262$ rather than exactly 2, and finite HRS leakage shifts every level, placing level 0.5 at 0.492. This is separate from readout error and is currently absorbed into the reported digital ceiling. It is correctable by trimming R_{MID} to $1082.9\text{ }\Omega$ or by folding the measured ratio into the block scale.

Scope. Results are ResNet-18 on CIFAR-10 and CIFAR-100 (FP2 QAT: 72.35% against 73.60% FP32). ImageNet is not yet evaluated.

X. CONCLUSION

Block-floating-point block size and CIM crossbar tile height are one physical parameter, not two. The resulting collision degrades ResNet-18 to chance at the tile heights that make a crossbar efficient. The mechanism is a shared-sense-resistor loading gain that is deterministic in the programmed conductances and therefore cancellable exactly by one compile-time constant per column, applied per bitline before differential

subtraction. Cancelling it converts tile height from a binding constraint into a free design parameter, yielding $+14.77$ accuracy points and $13.1\times$ energy efficiency simultaneously, verified against exhaustive SPICE and robust to 20% device variability. Conductance drift requires periodic recalibration, which is cheap and logarithmically infrequent.

More broadly, the failure mode reported here is invisible to additive-Gaussian analog error models, which remain standard in CIM evaluation. Systematic, data-independent gains warrant circuit-exact treatment.

ACKNOWLEDGMENT

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